

MAE 3780 ROBOT PROJECT FINAL REPORT

“FAAFO”

1. Robot Design and Strategy Overview.



Fig.1: Faafo, team 17's robot, front view

The driving factor behind FAAFO's strategy is speed, simplicity, and to not be impacted or depend upon the strategy chosen by the opponent. We prioritized having simple and straightforward mechanisms and code, to improve our reliability and reduce the risk of failure during competition.

The robot's path during competition was hardcoded and was designed to let us quickly reach the center of the board, collect a majority of the blocks, and then move to a safer and more secluded part of the board to hide from the other robot and avoid any interference. Specifically, our robot's strategy was to begin located at the center of our board, drive forward at a slight diagonal to the center (to try to avoid robots that drove directly forward

in the center), and then turn to the left to align with the center line. The robot then drove forward to the boundary of the board while collecting cubes, before turning to the right and driving forward to stop at the end of the board on the opponent's side.

The same simplicity-focused mindset was applied to the design of the arms. They are lightweight, since they are made of cardboard, and are fixed to the robot via loose hinges at a vertical angle so that they are calibrated to fall as soon as the robot starts moving. To help the arms fall forward, a very short backwards motion was added at the beginning of our code.

Another main aspect that was taken into consideration, was the storage of the cubes after collection. For that purpose, and to achieve a slightly higher robot speed, we designed 3D-printed extensions made to be clipped directly onto the provided wheels. This allowed us to create a protected storage space underneath our robot. To prevent cubes from interfering with the wheels and the rolling ball, we added a "cage" component underneath our chassis, made from cardboard, which both helped contain the cubes and prevented them from running into the wheels. (See pictures Appendix F for views from the underneath and back of FAAFO).

2. Design Process Reflection.

Through our design cycle, our group initially focused on passing the class milestones before turning to design and test our final competitive robot. Due to our existing class code, we progressed smoothly through the milestones and completed them well ahead of their due dates. We created components for our robot as necessary to pass each milestone, such as designing different circuits or making temporary arms to collect blocks. Once completing the milestones, we transitioned to focusing solely on the design, fabrication, and testing of our robot.

We began our design process by meeting and brainstorming different strategies and bot ideas. After landing on our desired design, we made multiple sketches and diagrams and began building a prototype to test the feasibility of our idea and get a better understanding of what additions/changes we needed to make. We mainly used robot parts provided to us in our kits, including the electronics, chassis, and drivetrain system.

After building the first iteration of our robot and validating the concept of our strategy, we found that we needed to increase the diameter of our wheels, rotate the position of our Arduino and breadboard (to allow more space for the arms), and create finalized versions of all of our cardboard components including the arms and storage compartment. Additionally with our design finalized, we ordered hinges and decorative design elements from McMaster. With this information, we began working on a CAD file for our 3D printed wheel extensions and cutting out the cardboard pieces. For pieces like the cardboard arm, we focused on keeping them at a length that were within the dimension constraints of the competition. Once those parts were finalized, we attached them to our assembly and completed a successful test drive with all of our components.

After finishing the fabrication and assembly of our robot, we began the process of coding. Due to the simplicity of our strategy and the lack of complex components like sensors, we were able to reuse our code from Milestone 2. The most challenging part was doing iterative testing to perfectly hard code the robot's path to be as we wanted to ensure it would perform as wished and reliably in competition. Ultimately, our final product ended up being very close to our original design.

Finally, we decorated our robot to match our overall purple alien theme. We encountered a small issue when some spray paint got in the crevices of our hinges and restricted their movement. We were able to mostly fix the issue by using an x-acto knife to scrape off the paint and loosen the hinge.

3. Competition Analysis.

Our robot performed very well during competition day, ultimately making it to the quarter finals. The storage compartment underneath our robot was very effective, and helped to protect our cubes from being stolen from opposing robots or falling out if our robot got knocked around. Similarly, our wheel extensions proved to be highly beneficial allowing the blocks to pass under our chassis and slightly improving our speed. Additionally, our strategy of hard coding the path of our robot proved to be relatively successful and reliable, allowing us to collect many blocks before attempting to avoid interactions with the opponent.

Although we only lost two rounds, both losses were caused by the same issue: our robot would occasionally get hooked by another robot and partially flipped or tipped onto its side, which would stop our ability to drive and cause us to lose control of our cubes. Additionally, our cardboard arm design presented some advantages and disadvantages. While their flexibility sometimes allowed us to avoid getting trapped or hooked on another robot, it also sometimes caused us to lose some cubes.

Overall, our robot consistently collected a large number of cubes and won most of our matches/ This validated the effectiveness of our overall strategy, and showed that our design could perform reliably and competitively through the competition.

4. Conclusions.

Our robot performed reliably throughout the competition, reaching the quarter finals and winning the majority of its matches. Our strategy of prioritizing speed and simplicity proved effective, as the hardcoded path consistently allowed us to collect a large number of cubes. The wheel extensions and cardboard storage cage were among our most successful design choices, protecting collected cubes and improving speed.

If we could redo this project, we would reinforce the robot's structural stability to prevent tipping, our only two losses stemmed from being hooked by opposing robots. We would also explore more rigid arm materials to reduce cube loss while maintaining flexibility.

For future students, keep your strategy simple, reliability outweighs complexity. Invest time in iterative path testing, as small deviations in hardcoded timing can significantly affect performance. Testing against physical opponents before competition day is essential, as it reveals weaknesses that solo testing cannot.

5. Appendix A: Bill of Materials (BOM).

Product	Units	Price	Weight
Painter Tape	1	\$3.72	Negligible
Purple Spray Paint	1	\$4.48	N/A
Wheel extension	2	approx. \$5.00	~ 50g
Hinges (x2)	1	\$2.20	~ 10g
TOTAL	5	\$22.60	

6. Appendix B: circuit diagram.

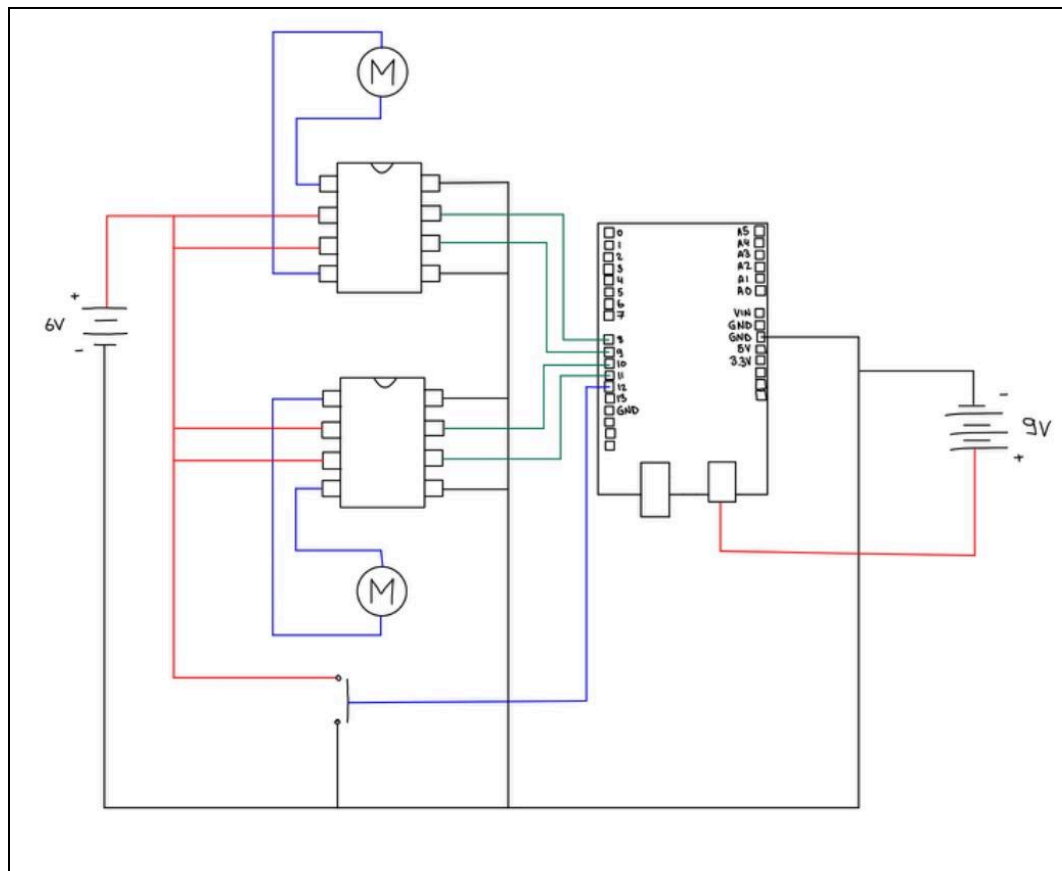


Fig.2: Circuit Diagram

7. Appendix C: CAD files and drawings.



Fig.3: CAD of wheel extension

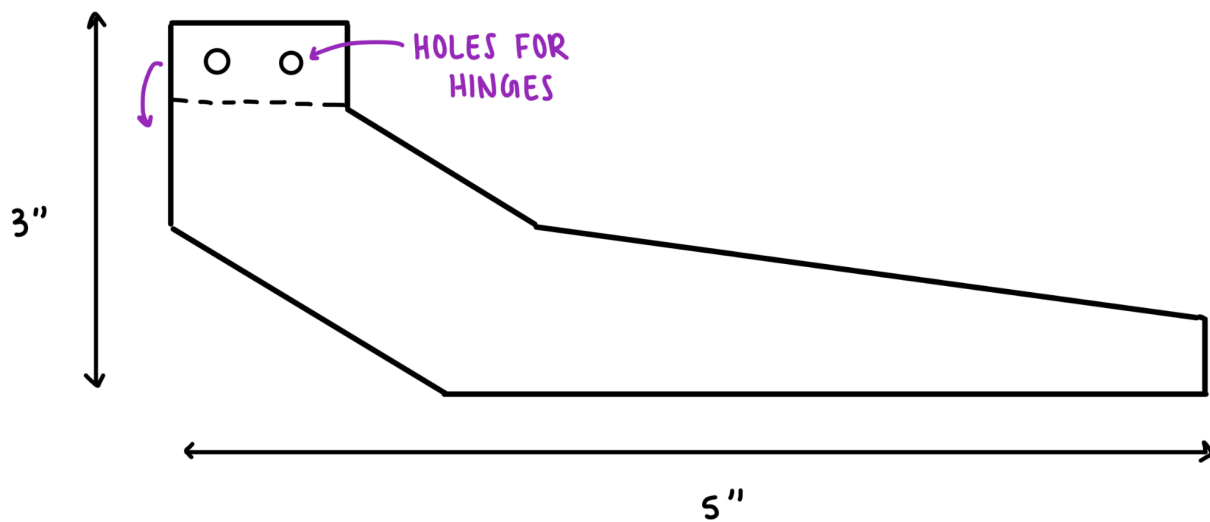


Fig.4: Cardboard Arm Schematic

8. Appendix D: Flowchart.

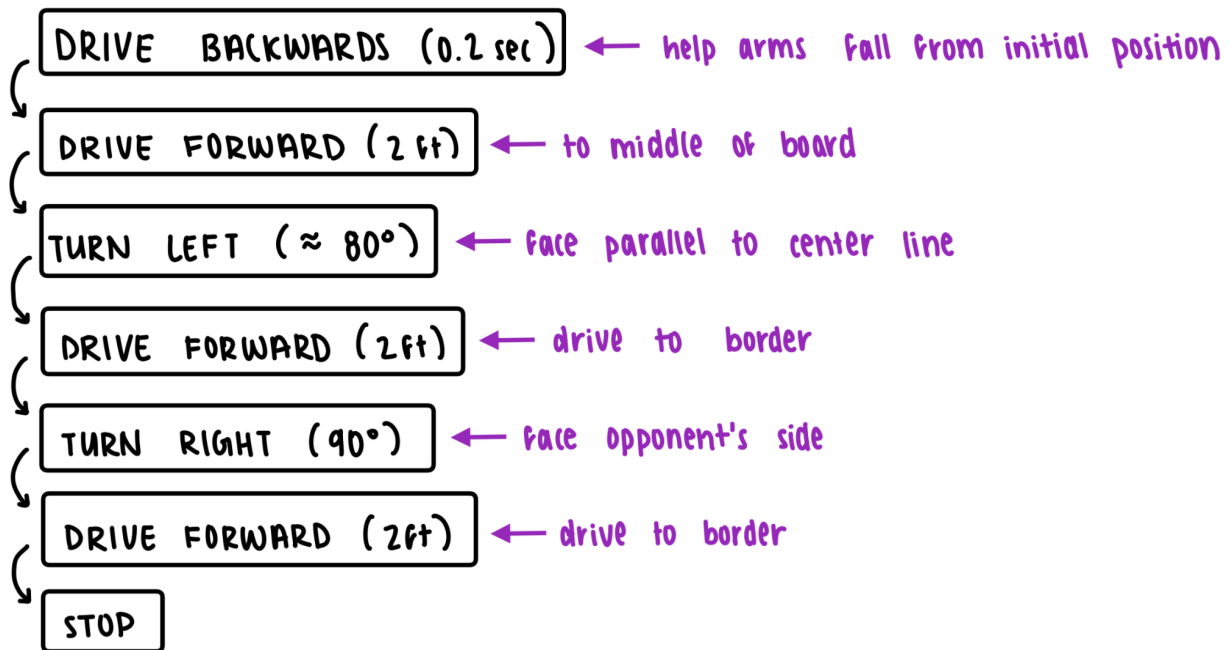


Fig.5: Flowchart (read top to bottom)

9. Appendix E: Code: commented Arduino code.

```
#include <avr/io.h>
#include <util/delay.h>

int main(void){ // setup code that only runs once
  DDRB |= 0b00000011;// set pins 8 and 9 as outputs for "right" motor
  DDRB |= 0b00001100;// set pins 10 and 11 as outputs for "left" motor

  while(1){ // code that loops forever
    if (PINB & 0b00010000){ // when switch on, follow pattern by calling functions
      drive_forward(200);
      drive_backward(2600);
      drive_left(500);
      drive_backward(2200);
      drive_right(550);
      drive_backward(2500);
      stop(500000);
    }
    else {
    }
  }
}
```

```
// delay function and moving/stopping functions
```

```
void delay_ms(int ms) {  
    while (ms-->0) {  
        _delay_ms(1);  
    }  
}
```

```
void drive_forward(int t){  
    PORTB |= 0b00001001;  
    delay_ms(t);  
    PORTB &=~ 0b00001001;  
}
```

```
void drive_backward(int t){  
    PORTB |= 0b00000110;  
    delay_ms(t);  
    PORTB &=~ 0b00000110;  
}
```

```
void drive_left(int t){  
    PORTB |= 0b00000101;  
    delay_ms(t);  
    PORTB &=~ 0b00000101;  
}
```

```
void drive_right(int t){  
    PORTB |= 0b00001010;  
    delay_ms(t);  
    PORTB &=~ 0b00001010;  
}
```

```
void stop(int t){  
    PORTB &=~ 0b00001111;  
    delay_ms(t); }
```

10. Appendix F: Pictures



Fig.6: Views of Faafo from different angles